Chiral relations for entanglement in quantum self-phase modulation

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We investigate chiral selection rules for bipartite and multipartite entanglement in the radial mode structure generated in self phase modulation with orbital angular momentum (OAM). While OAM conservation is a trivial selection rule for entanglement, a less intuitive rule applies to the radial structure. Depending on the relative chirality of the interacting modes, entanglement can be concentrated between two radial modes or distributed in a multiparty configuration determined by the OAM values involved. This opens the possibility of generating and controlling multipartite entanglement in the radial structure.

I. INTRODUCTION

The quest for quantum devices based on optical resources is an intense and active field of research, which has already delivered important tools for quantum communication, computing and metrology. This quest has increased considerably by the possibility of shaping light in its different degrees of freedom, such as time envelope, spectral and spatial modes [1, 2]. Both discrete and continuous variable quantum information processing can benefit from these developments [3–5]. Nonlinear optical processes are the common tool for generating entangled quantum states, as it allows the energy transfer between different optical modes, constrained by different conditions such as energy conservation and phase matching, for example [6–10]. These conditions are the basis for entanglement, as they impose different kinds of quantum correlations among the modes interacting in the nonlinear process [11–14]. Energy and momentum conservation are behind the generation of time-frequency and position-momentum entanglement, respectively. These conditions rule nonlinear optical processes such as three- and four-wave mixing, second harmonic generation and self-phase modulation[15–25].

The possibility of shaping the phase and polarization distributions with spatial light modulators and q-plates has become a powerful tool for controlled generation of customized entangled states both in linear and nonlinear optics. These developments brought the orbital angular momentum (OAM) of light to the toolbox of quantum information technology. Its role in nonlinear processes has also been unveiled by a plethora of scientific papers devoted to the constraints imposed by OAM and vortex formation in nonlinear wave mixing. In polariton superfluids, the interplay between phase, spin and flow gives rise to a rich variety of spatial structures, including quantized vortices, spin textures and vortex flows [26–30]. A recent work has further demonstrated the spectroscopy of collective excitations associated with stationary, multiply quantized vortices, opening new possibilities for exploring vortex dynamics in driven-dissipative quantum fluids of light [30].

A less exploited degree of freedom involves the radial structure of paraxial modes [8, 15, 16, 31–33]. The Laguerre-Gaussian (LG) solutions of the paraxial wave equation are the natural basis for describing OAM and the radial structure, which are the suitable labels for their identification. These modes are characterized by two integer numbers, namely its topological charge and radial index. The topological charge corresponds to the azimuthal variation of the beam phase around the propagation direction and the radial index determines the number of intensity rings. While the OAM conservation selection rule has been demonstrated in different quantum optical processes, the constraints related to the radial structure still deserves careful investigation.

Self-phase modulation is a particularly interesting phenomenon, where the intensity of an optical beam alters the effective refraction index of a nonlinear medium, thus affecting the beam's phase evolution along propagation through the medium [34]. This effect can also be observed in Bose-Einstein condensates and atomic vapors [35–37]. In a previous work we investigated the formation of a radial structure in self-phase modulation driven by an OAM beam and determined the radial spectrum generated in terms of the topological charge of the input beam. This established an interesting radial-angular coupling promoted by the radial selection rules imposed on the nonlinear process [38].

In the present work we investigate the consequence of radial-angular coupling and the associated selection rule to entanglement generation. In particular, we will discuss bipartite and multipartite entanglement generated in the process, which can server as a possible platform for quantum information protocols based on these resources. This platform is particularly interesting due to the wide range of experimental tools for its implementation, going from polariton superfluids to atomic vapors.

II. SPATIALLY STRUCTURED BOSONIC FIELD

Before addressing the effective Hamiltonian of an optical field undergoing nonlinear propagation in a Kerr medium, let us establish the formal background allowing for the treatment of quantized spatial modes and their relevant quantum properties, including entanglement. Although the quantum theory of spatial optical modes have been extensively discussed in the literature, we would like to set the relevant concepts and definitions that will be used when dealing with the quantized mode structure of the optical field.

We start by considering the normalized positive frequency component of the quantized electromagnetic field $\mathcal{E}(\mathbf{r})$ satisfying the commutation relations

$$\begin{aligned}\left[\mathcal{E}(\mathbf{r}_1),\mathcal{E}^\dagger(\mathbf{r}_2)\right] &= \delta^{(2)}(\mathbf{r}_1-\mathbf{r}_2),\\ \left[\mathcal{E}(\mathbf{r}_1),\mathcal{E}(\mathbf{r}_2)\right] &= \left[\mathcal{E}^\dagger(\mathbf{r}_1),\mathcal{E}^\dagger(\mathbf{r}_2)\right] = 0.\end{aligned} \tag{1}$$

In the above equation and in what follows, $\mathbf{r} = (x, y)$ is the transverse coordinate. One can introduce ladder operators $a(v), a^\dagger(v)$ associated with the spatial mode v by the expression

$$a(v) = \int_{\mathbb{R}^2} v^*(\mathbf{r})\mathcal{E}(\mathbf{r})d\mathbf{r}. \tag{2}$$

We then have

$$\begin{aligned}\left[a(v_1),a^\dagger(v_2)\right] &= (v_1;v_2) \equiv \int_{\mathbb{R}^2} v_1^*(\mathbf{r})v_2(\mathbf{r})d\mathbf{r}\,,\\ \left[a(v_1),a(v_2)\right] &= \left[a^\dagger(v_1),a^\dagger(v_2)\right] = 0\,.\end{aligned} \tag{3}$$

Note that this construction allows us to deal with partially overlapping or non normalized modes, reducing to the standard form when $(v_1; v_2) = 0$ or $(v_1; v_2) = 1$ $(v_1 = v_2)$. This feature will be useful when dealing with bi- and multipartite entanglement among the modes coupled in the nonlinear medium.

Given a classical field distribution $\alpha(\mathbf{r})$ normalized in such a way that $|\alpha(\mathbf{r})|^2$ represents the photon number per unit area, we may build a coherent state through the expression

$$|\alpha\rangle = \exp\left[a^\dagger(\alpha) - a(\alpha)\right]|0\rangle\ , \tag{4}$$

where $|0\rangle$ is the vacuum state. This state is an eigenvector of the field operator,

$$\mathcal{E}(\mathbf{r})\,|\alpha\rangle = \alpha(\mathbf{r})\,|\alpha\rangle\ . \tag{5}$$

This is the type of state that will be used as input on the nonlinear medium.

The simplest entanglement criteria for bosonic systems are based on the field quadratures, which may be defined through the formula

$$X(v) = \frac{a(v) + a^\dagger(v)}{\sqrt{2}}\,. \tag{6}$$

Here, v is the corresponding field mode. The complementary quadrature is simply defined by shifting v by a $\pi/2$ phase: $P(v) = X(iv)$.

The corresponding covariance function is given by

$$\Gamma(v_1, v_2) = \langle\{\Delta X(v_1), \Delta X(v_2)\}\rangle\,, \tag{7}$$

where $\Delta X(v) = X(v) - \langle X(v)\rangle$ and $\{\bullet,\bullet\}$ denotes the anticommutator. The calculation of this observable is simplified if we express it in terms of the field $\mathcal{E}$. For this, we introduce the normal correlation

$$H(\mathbf{r}_1,\mathbf{r}_2) = \left\langle\mathcal{E}^\dagger(\mathbf{r}_2)\mathcal{E}(\mathbf{r}_1)\right\rangle - \left\langle\mathcal{E}^\dagger(\mathbf{r}_2)\right\rangle\langle\mathcal{E}(\mathbf{r}_1)\rangle\ , \tag{8}$$

and the anomalous one

$$F(\mathbf{r}_1,\mathbf{r}_2) = \langle\mathcal{E}(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z)\rangle - \langle\mathcal{E}(\mathbf{r}_1)\rangle\langle\mathcal{E}(\mathbf{r}_2)\rangle\,. \tag{9}$$

We may then write

$$\Gamma(v_1,v_2) = \mathrm{Re}\left[(v_1;v_2) + 2f(v_1,v_2) + 2h(v_1,v_2)\right], \tag{10}$$

where

$$\begin{aligned}h(v_1,v_2) &= \int v_1^*(\mathbf{r}_2)v_2(\mathbf{r}_1)H(\mathbf{r}_1,\mathbf{r}_2)\,d\mathbf{r}_1\,d\mathbf{r}_2\\ &= \left\langle a^\dagger(v_2)a(v_1)\right\rangle - \left\langle a(v_2)\right\rangle^*\langle a(v_1)\rangle\end{aligned} \tag{11}$$

and

$$\begin{aligned}f(v_1,v_2) &= \int v_1^*(\mathbf{r}_2)v_2^*(\mathbf{r}_1)F(\mathbf{r}_1,\mathbf{r}_2)\,d\mathbf{r}_1\,d\mathbf{r}_2\\ &= \langle a(v_1)a(v_2)\rangle - \langle a(v_1)\rangle\langle a(v_2)\rangle\end{aligned} \tag{12}$$

We further note the following properties, which simplify the consideration of linear combination of modes:

$$\begin{aligned}f(v_1,v_2+\lambda v_3) &= f(v_1,v_2) + \lambda^* f(v_1,v_3)\\ f(v_1,v_2) &= f(v_2,v_1)\\ h(v_1,v_2+\lambda v_3) &= h(v_1,v_2) + \lambda h(v_1,v_3)\\ h(v_1,v_2) &= h(v_2,v_1)^*\end{aligned} \tag{13}$$

These are a reflection of the fact that $a(v)$ is antilinear while $a^\dagger(v)$ is linear.

A. Single-mode correlations

Let us analyze the covariance matrix $\gamma_{jk} = \langle\{\Delta Z_j, \Delta Z_k\}\rangle$ where $\mathbf{Z} = (X(v), P(v))$. With our quadrature normalization, the vacuum variance is 1/2. We therefore express quadrature variances in shot-noise units (SNU), defining $V_{\mathrm{SNU}}(Q) = 2\left\langle(\Delta Q)^2\right\rangle$, so that the vacuum variance is one and the eigenvalues of γ are directly the quadrature variances in SNU. We observe that $h(v, v) \in \mathbb{R}$, so we get

$$\gamma = (1 + 2h(v,v))\,\mathbb{1}_2 + 2\,\mathrm{Re}\,f(v,v)\sigma_z + 2\,\mathrm{Im}\,f(v,v)\sigma_x\ , \tag{14}$$

where $\mathbb{1}_2$ is the 2×2 identity, and σ_x and σ_z are the corresponding Pauli matrices. Notice that $f(e^{i\phi}v, e^{i\phi}v) = e^{-i2\phi}f(v,v)$. Therefore, we can diagonalize γ by choosing $\phi = [\arg f(v,v) \pm \pi]/2$ and obtain the eigenvalues

$$\lambda_\pm = 1 + 2\,[\,h(v,v) \pm |f(v,v)|\,]\ . \tag{15}$$

We then see that the quadrature rotated by ϕ has minimum variance λ_- in SNU, while the one rotated by $\phi + \pi/2$ has maximum variance λ_+. Furthermore, we have squeezing if $|f(v,v)| > h(v,v)$.

B. Duan Criterion for two-mode entanglement

Besides the internal correlations characterizing squeezing or excess noise within each individual mode, we wish to describe inter mode correlations and entanglement. Consider two orthonormal states v_1, v_2, i.e., $(v_j, v_k) = \delta_{jk}$. The Duan criterion [39] states that if

$$\begin{aligned} D = &\frac{1}{2}\left\langle [\Delta X(v_1) + \Delta X(v_2)]^2 \right\rangle \\ &+ \frac{1}{2}\left\langle [\Delta P(v_1) - \Delta P(v_2)]^2 \right\rangle < 1\ . \end{aligned} \tag{16}$$

then the states are entangled.

Let us define new modes $v_\pm = (v_1 \pm v_2)/\sqrt{2}$, so that

$$D = \left\langle \Delta X(v_+)^2 \right\rangle + \left\langle \Delta P(v_-)^2 \right\rangle\ . \tag{17}$$

By applying (10) we obtain

$$\begin{aligned} D &= 1 + h(v_+, v_+) + h(v_-, v_-) \\ &\quad + \mathrm{Re}\,[f(v_+, v_+) - f(v_-, v_-)] \\ &= 1 + h(v_1, v_1) + h(v_2, v_2) + 2\,\mathrm{Re}\, f(v_1, v_2)\ . \end{aligned} \tag{18}$$

Once again, we can maximize the possible violation by rotating the quadrature of one of the modes, say, v_1 by an angle $\phi = \arg f(v_1, v_2) \pm \pi$, which gives the optimal value

$$D_{\rm opt} = 1 + h(v_1, v_1) + h(v_2, v_2) - 2|f(v_1, v_2)|\ . \tag{19}$$

III. THE KERR SYSTEM

We now turn to the quantum description of the Kerr dynamics. Let us consider a monochromatic light field propagating through a $\chi^{(3)}$ nonlinear medium, undergoing self-phase modulation. The input field is assumed to be initially prepared in a coherent state of a spatial mode. Besides self-phase modulation, the input mode is coupled to the vacuum fluctuations of other modes, sharing energy and information through the nonlinear interaction. As we will see, this creates a rich dynamics where energy and information exchange is subjected to selection rules that impose a chiral relation that couples the radial and angular degrees of freedom of the interacting modes and determine the formation of entangled clusters.

A. Heisenberg Equation for the Field and Linear Solution

We will assume that the field is propagating along the z direction, and satisfies the Heisenberg equation of motion

$$2ik\partial_z \mathcal{E}(\mathbf{r}, z) + \nabla^2 \mathcal{E}(\mathbf{r}, z) = g\,\mathcal{E}^\dagger(\mathbf{r}, z)\mathcal{E}^2(\mathbf{r}, z)\,, \tag{20}$$

where g is the Kerr nonlinear coupling constant and ∇^2 denotes the laplacian in the transverse direction. We consider will two distinct approximations starting from the same field decomposition: a short-distance expansion retaining the full fluctuation equation, and a diffraction-free, linearized fluctuation treatment without a Taylor expansion in distance.

To start our derivation, let $\alpha_c(\mathbf{r}, z)$ denote the solution of the classical equations of motion

$$2ik\partial_z \alpha_c + \nabla^2 \alpha_c = g\,|\alpha_c|^2 \alpha_c \tag{21}$$

under the initial condition $\alpha_c(\mathbf{r}, 0) = \alpha(\mathbf{r})$, where $|\alpha\rangle$ is the input coherent state. Let us also define the fluctuation field

$$\eta(\mathbf{r}, z) = \mathcal{E}(\mathbf{r}, z) - \alpha_c(\mathbf{r}, z) \tag{22}$$

We observe that our initial state has vacuum like properties with respect to $\eta(\mathbf{r}, 0)$. For example, it is annihilated by it:

$$\eta(\mathbf{r}, 0)\,|\alpha\rangle = 0 \tag{23}$$

Furthermore, any normally ordered monomial containing at least one fluctuation operator has zero expectation value at $z = 0$.

The correlation functions, which are our observables of interest, can also be compactly written in terms of η:

$$\begin{aligned} F(\mathbf{r}_1, \mathbf{r}_2; z) &= \langle \eta(\mathbf{r}_1, z)\eta(\mathbf{r}_2, z)\rangle - \langle \eta(\mathbf{r}_1, z)\rangle\langle \eta(\mathbf{r}_2, z)\rangle, \\ H(\mathbf{r}_1, \mathbf{r}_2; z) &= \langle \eta^\dagger(\mathbf{r}_2, z)\eta(\mathbf{r}_1, z)\rangle - \langle \eta(\mathbf{r}_1, z)\rangle\langle \eta^\dagger(\mathbf{r}_2, z)\rangle. \end{aligned} \tag{24}$$

Next, we write the equation of motion

$$2ik\partial_z \eta + \nabla^2 \eta = g\left(2|\alpha_c|^2\eta + \alpha_c^2\eta^\dagger + R\right) \tag{25}$$

where the nonlinear fluctuation terms are

$$R = \alpha_c^*\eta^2 + 2\alpha_c\eta^\dagger\eta + \eta^\dagger\eta^2. \tag{26}$$

We start by considering an approximation where only terms up to $\mathcal{O}(z)$ are retained. From definition, we already have $\langle \eta(\mathbf{r}, 0)\rangle = 0$. Furthermore, as the equation of motion (25) contains only normally ordered terms, we also get $\partial_z \langle \eta(\mathbf{r}, 0)\rangle = 0$, such that

$$\langle \eta(\mathbf{r}, z)\rangle = \mathcal{O}(z^2), \tag{27}$$

or, equivalently,

$$\langle \mathcal{E}(\mathbf{r}, z)\rangle = \alpha_c(\mathbf{r}, z) + \mathcal{O}(z^2). \tag{28}$$

Now we turn to the correlation functions. For compactness, we write $\eta_{j,z} = \eta(\mathbf{r}_j, z)$. In order to calculate the normal correlation, we differentiate the two-operator average:

$$\partial_z \langle \eta_2^\dagger \eta_1 \rangle \Big|_0 = \langle (\partial_z \eta_2^\dagger)_0 \eta_{0,1} \rangle + \langle \eta_{0,2}^\dagger (\partial_z \eta_1)_0 \rangle. \tag{29}$$

Both terms vanish: the first average has a vacuum-annihilating operator on the right, and the second has one on the left.

Turning, to the anomalous case, we have that

$$\begin{aligned} \partial_z \langle \eta_{1,z} \eta_{2,z} \rangle|_0 &= \langle (\partial_z \eta_1)_0 \eta_{2,0} \rangle + \langle \eta_{1,0} (\partial_z \eta_2)_0 \rangle \\ &= \langle \eta_{1,0} (\partial_z \eta_2)_0 \rangle. \end{aligned} \tag{30}$$

Isolating $\partial_z \eta$ in the equations of motion (25) and substituting above, we see that almost all the terms are zero, because they contain η to the right. The only exception is the term proportional to $\eta^\dagger$, which gives

$$\partial_z \langle \eta_1 \eta_2 \rangle|_0 = -\frac{ig}{2k} \alpha^2(\mathbf{r}_2) \delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2). \tag{31}$$

Since the mean-subtraction terms do not contribute at first order, the correlations are

$$\begin{aligned} F(\mathbf{r}_1, \mathbf{r}_2; z) &= -\frac{igz}{2k} \alpha^2(\mathbf{r}_2) \delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2) + O(z^2), \\ H(\mathbf{r}_1, \mathbf{r}_2; z) &= O(z^2). \end{aligned} \tag{32}$$

Using the spatial overlap notation

$$(v_1, \ldots, v_n; u_1, \ldots, u_n) = \int d^2\mathbf{r} \prod_{j=1}^{n} v_j^*(\mathbf{r}) u_j(\mathbf{r}), \tag{33}$$

projection onto the modes v_1 and v_2 gives

$$\begin{aligned} f(v_1, v_2) &= -\frac{igz}{2k} (v_1, v_2; \alpha, \alpha) + O(z^2), \\ h(v_1, v_2) &= O(z^2). \end{aligned} \tag{34}$$

B. Laguerre-Gaussian inputs and dimensionless form

We will consider coherent Laguerre-Gaussian modes as our input states, with

$$\alpha(\mathbf{r}, 0) = \|\alpha\| \, LG_{pl}(\mathbf{r}; w), \tag{35}$$

where the waist-plane modes are

$$LG_{pl}(\mathbf{r}; w) = (-1)^p \frac{\mathcal{N}_{pl}}{w} \left(\frac{2r^2}{w^2} \right)^{|l|/2} L_p^{|l|} \left(\frac{2r^2}{w^2} \right) e^{-r^2/w^2 + il\phi}, \tag{36}$$

with radial order p, topological charge l, and normalization constant

$$\mathcal{N}_{pl} = \sqrt{\frac{2}{\pi}} \sqrt{\frac{p!}{(p + |l|)!}} \, . \tag{37}$$

The waist w sets the natural transverse scale. Let us define the Rayleigh range $z_R = kw^2/2$, the nonlinear interaction length $z_{NL} = 4z_R/(g\|\alpha\|^2)$, the dimensionless propagation distance $Z = z/z_{NL} = gz\|\alpha\|^2/(2kw^2)$, and the dimensionless input profile $u(\mathbf{r}) = w\alpha(\mathbf{r}, 0)/\|\alpha\|$.

Using these definitions, the first-order results take the dimensionless form

$$\begin{aligned} f(v_1, v_2) &\approx -iZ(v_1, v_2; u, u), \\ \Gamma(v_1, v_2) &= \mathrm{Re}(v_1; v_2) + 2Z \, \mathrm{Im}(v_1, v_2; u, u), \\ \lambda_- &= 1 - 2\,|Z(v, v; u, u)|\,, \\ D_{\mathrm{opt}} &= 1 - 2\,|Z(v_1, v_2; u, u)|\,, \end{aligned} \tag{38}$$

For the input angular momentum l_0, we see that the angular integration requires $l_v = l_0$ for squeezing and $l_{v_1} + l_{v_2} = 2l_0$ for entanglement. These OAM conditions are necessary; the corresponding radial overlap must also be nonzero.

C. Diffraction-free Bogoliubov solution

The first-order expressions describe the initial generation of correlations, but extrapolating them to larger distances can give negative values of λ_- or D_{opt}. For finite-distance predictions, we return to the full equation for the same decomposition $\mathcal{E} = \alpha_c + \eta$ and adopt a different approximation: we neglect diffraction and discard the nonlinear fluctuation terms R in Eq. (25), while retaining the full dependence on propagation distance. This linearization of the fluctuation dynamics is the Bogoliubov approximation, used to describe quantum fluctuations in paraxial propagation through Kerr media [40]. Appendix A solves the diffraction-free Bogoliubov equations without a Taylor expansion in distance. The dimensionless classical field is then

$$u_Z(\mathbf{r}) = u(\mathbf{r}) e^{-iZ|u(\mathbf{r})|^2}, \tag{39}$$

and the mode correlations are

$$\begin{aligned} f(v_1, v_2) = & -iZ(v_1, v_2; u_Z, u_Z) \\ & - Z^2(v_1, v_2, u_Z; u_Z, u_Z, u_Z), \\ h(v_1, v_2) = & \, Z^2(v_1, u_Z, u_Z; v_2, u_Z, u_Z). \end{aligned} \tag{40}$$

These expressions agree with the preceding first-order result and are exact within the diffraction-free linearized fluctuation equations. Their use at finite distance requires negligible diffraction over the interaction region, which is a reasonable approximation for softly focused beams, as well as a small fluctuation backreaction. The numerical results below use these correlations under those assumptions.

IV. SELECTION RULES FOR LAGUERRE-GAUSSIAN INPUTS

We now specialize to the LG input $\alpha(\mathbf{r}, 0) = \|\alpha\| \, LG_{0,l_0}(\mathbf{r}; w)$ and analyze the selection rules for ra-

dial and angular mode coupling.

Our previous work [38] has shown that, in the classical regime, the first order treatment gives a selection rule for the generation of radial orders up to $|l_0|$. The current quantum treatment clarifies that this was, in fact, a selection rule for the mean field value $\langle\mathcal{E}\rangle$ given by (28), whose expression had already appeared on our earlier treatment.

Now that we have developed a quantum theory for the considered process, we may analyze other types of selection rules, this time for bi- and multipartite entanglement. Beyond the obvious conservation law, orbital angular momentum also affects radial mode coupling through chiral relations between the interacting modes, as we will see in the remaining of this section.

A. Two mode entanglement

Consider the case where Furthermore, let $v_2 = LG_{0,l_2}(\mathbf{r}; w_2)$ where w_2 is free and l_2 is only constrained by $l_2 \neq l_0$. Now, define $l_1 = 2l_0 - l_2$ and $1/w_1^2 = 2/w^2 + 1/w_2^2$. We then have that

$$\begin{aligned} v_2^*(\mathbf{r})u^2(\mathbf{r}) &= \frac{\mathcal{N}_{0l_2}\mathcal{N}_{0l_0}^2}{w_2}\left(\frac{2r^2}{w_2^2}\right)^{\frac{|l_2|}{2}}\left(\frac{2r^2}{w^2}\right)^{|l_0|} e^{-\frac{r^2}{w_1^2}+il_1\phi} \\ &= \frac{\mathcal{N}_{0l_2}\mathcal{N}_{0l_0}^2\, w_1^{2|l_0|+|l_2|}}{w_2^{|l_2|+1}w^{2|l_0|}}\left(\frac{2r^2}{w_1^2}\right)^{\frac{|l_1|}{2}}\left(\frac{2r^2}{w_1^2}\right)^{p_{\max}} \\ &\quad\times e^{-\frac{r^2}{w_1^2}+il_1\phi}\,, \end{aligned} \tag{41}$$

where we have defined

$$\begin{aligned} p_{\max} &= |l_0| + \frac{|l_2| - |2l_0 - l_2|}{2} \\ &= \begin{cases} 0, & l_0 l_2 \leq 0 \\ \min(2|l_0|, |l_2|), & l_0 l_2 > 0 \end{cases}. \end{aligned} \tag{42}$$

Then, by expanding the monomial that is raised to the power of $p_{\max}$ in terms of Laguerre polynomials, we get

$$v_2^*(\mathbf{r})u^2(\mathbf{r}) = \sum_{p=0}^{p_{\max}} C_p\, LG_{pl_1}(\mathbf{r}; w_1)\,, \tag{43}$$

where

$$C_p = \frac{p_{\max}!\mathcal{N}_{0l_2}\mathcal{N}_{0l_0}^2\, w_1^{2|l_0|+|l_2|+1}}{\mathcal{N}_{pl_1} w_2^{|l_2|+1} w^{2|l_0|}}\binom{p_{\max}+|l_1|}{p_{\max}-p}\,. \tag{44}$$

Therefore, setting $v_1 = LG_{pl_1}(\mathbf{r}; w_1)$, which is orthogonal to v_2, and considering the two mode subsystem formed by v_1, v_2, we get an optimal Duan violation of

$$D_{\text{opt}} = \begin{cases} 1 - 2C_p|Z|, & p \leq p_{\max} \\ 1, & p > p_{\max} \end{cases}. \tag{45}$$

Therefore, for nonzero Z within the first-order regime, when $p \leq p_{\max}$ we have bipartite entanglement in this two-mode subsystem. This selection rule was derived only for the two-mode Duan criterion and does not, by itself, imply a multipartite selection rule.

In Fig. 1 we show the violation of the Duan criterion as a function of Z for a few different combinations of l_0 and l_2. The selection rules can be clearly observed here: in all cases, the modes with $p > p_{\max}$ fail to violate the entanglement criterion (which does not mean absence of entanglement).

B. Multipartite entanglement

The work published in Ref. [41] introduces a generalized class of entanglement witnesses $\mathcal{Z}$. For a specified separability class, a witness is a real, symmetric, positive semidefinite matrix satisfying $\mathrm{Tr}\,(\mathcal{Z}\gamma_{\text{sep}}) \geq 1$ for every covariance matrix γ_{sep} of a state in that class. Therefore, if $\mathrm{Tr}\,(\mathcal{Z}\gamma) < 1$, every state with covariance matrix γ lies outside the specified separability class. A value greater than or equal to one is inconclusive and does not establish separability. The Duan criterion is a special case of a witness $\mathcal{Z}$ such that

$$\mathcal{Z} = \frac{1}{4}\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}. \tag{46}$$

Multipartite entanglement is a subtle topic, because separability becomes more complicated once there are more than two parties. To handle this, [41] introduces two types of witnesses:

- *Witness for entanglement with respect to a given partition*

  Here, one fixes a partition and asks whether the state is separable according to this partition. There is a numerical routine called FullyWit that finds the optimal $\mathcal{Z}$ for a given covariance matrix, in the sense that the violation $1 - \mathrm{Tr}\,(\mathcal{Z}\gamma)$ is maximized. A positive violation rules out separability with respect to the selected partition.

- *Witness for genuine multipartite entanglement*

  Suppose that now we have many parties and we want to decide if the state is separable with respect to these parties. One could then check individually all of the bipartitions with the previous method. For example, if we have three parties A, B and C, one could check individually the bipartitions $AB|C$, $AC|B$ and $A|BC$. This is actually sufficient for pure states, but, for mixed ones, we could write a state as a mixture of biseparable ones, say $\rho = \frac{1}{3}\rho_{AB|C} + \frac{1}{3}\rho_{AC|B} + \frac{1}{3}\rho_{A|BC}$. The full state ρ may not be separable with respect to any bipartition, but it is still a mixture of separable states. We then say that a state is genuine multipartite entangled if it cannot be written as a convex sum of

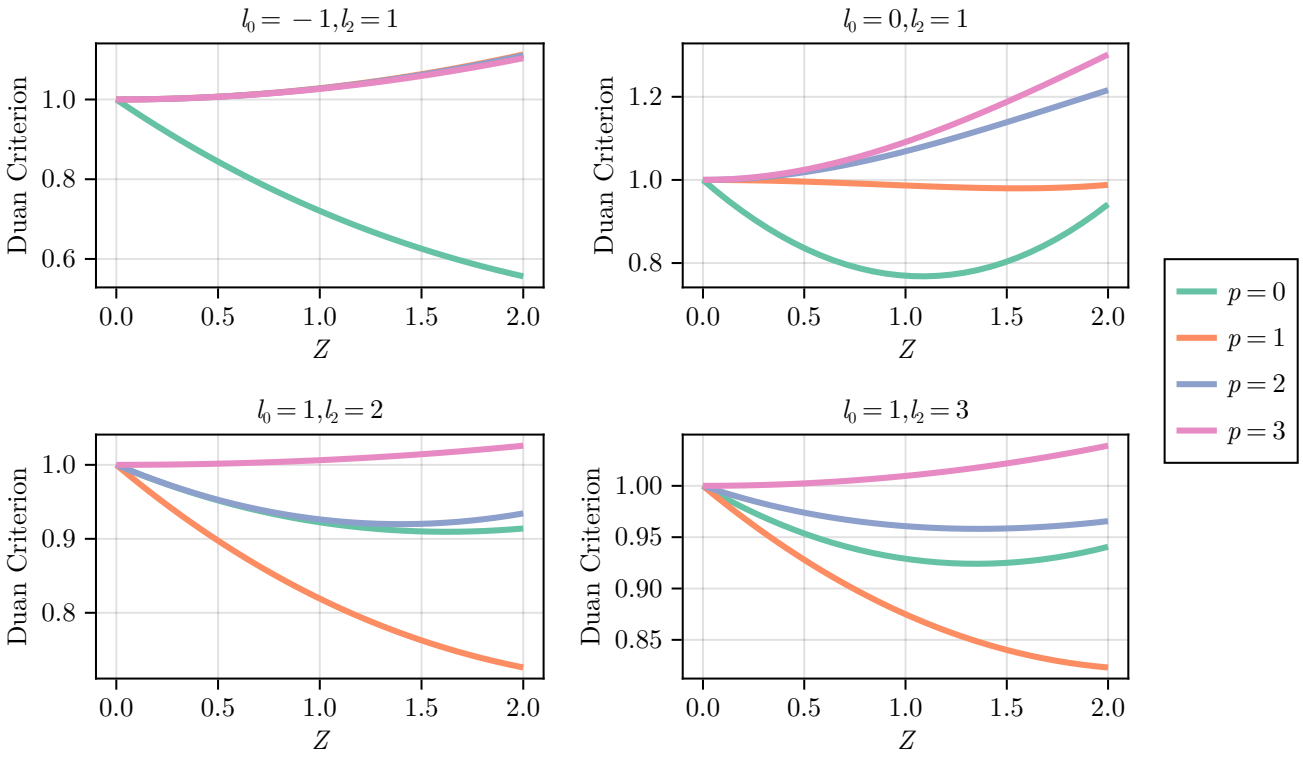


FIG. 1. Duan Criterion Value

bipartite separable states. The reference [41] also introduces a routine, MultiWit, that finds an optimized witness for genuine multipartite entanglement. A positive violation returned by this routine rules out the entire class of biseparable states and thus certifies genuine multipartite entanglement.

Now we employ these types of optimal entanglement witnesses for our Kerr system. In Fig. 2 we consider the same setup as in Fig. 1, but use the optimized witness. We do not see a large difference with respect to Fig. 1, which shows that the optimization over the angles used to arrive at (19) already carries us a long way.

Now we turn to more parties. We set $l_0 = 2$ and $l_2 = 1$, so that $l_1 = 2l_0 - l_2 = 3$ and $p_{max} = \min(2l_0, l_2) = 1$. Furthermore, we define the parties as $A = LG_{0,1}(w_2 = w)$, $B = LG_{0,3}(w_1)$, $C = LG_{1,3}(w_1)$ and $D = LG_{2,3}(w_1)$, so that D corresponds to a radial mode outside the set coupled by the radial-angular selection rule. According to the previous results, we only had entanglement between $A|B$ and $A|C$, while there were inconclusive results when considering $A|D$. In Fig. 3 we use the FullyWit routine to find optimized witnesses for all the bipartitions of these four modes. The lines seem to cluster in three groups

- $ACD|B$, $AC|BD$, $AD|BC$ and $A|BCD$ show the lowest witness values and hence the largest margins below the separability threshold. These share the fact that A and B belong to different parties, giving a stronger second-moment entanglement certification across these partitions.

- $ABD|C$ and $AB|CD$ show intermediate witness values, still maintaining a linear start. Here, A and B belong to the same party, while C is in the other one.

- $ABC|D$ shows the smallest margin below the separability threshold, starting with a quadratic fall. Here, all the modes that were bipartite entangled sit on the same party.

Finally, we turn to the genuine multipartite case. In Fig. 4, we show the witness for sets of the form $\{LG_{0,1}(w_2 = w)\} \cup \{LG_{p,3}(w_1) \mid p \leq P\}$, with P varying from 0 to 3, while maintaining $l_0 = 2$. Although no multipartite selection rule has been derived, the numerical results display a related pattern: the sets with $P \leq p_{\max} = 1$ have lower witness values and hence larger margins below the separability threshold.

V. CONCLUSION

We studied bi- and multipartite entanglement properties in the radial structure of the modes generated by self-phase modulation with orbital angular momentum

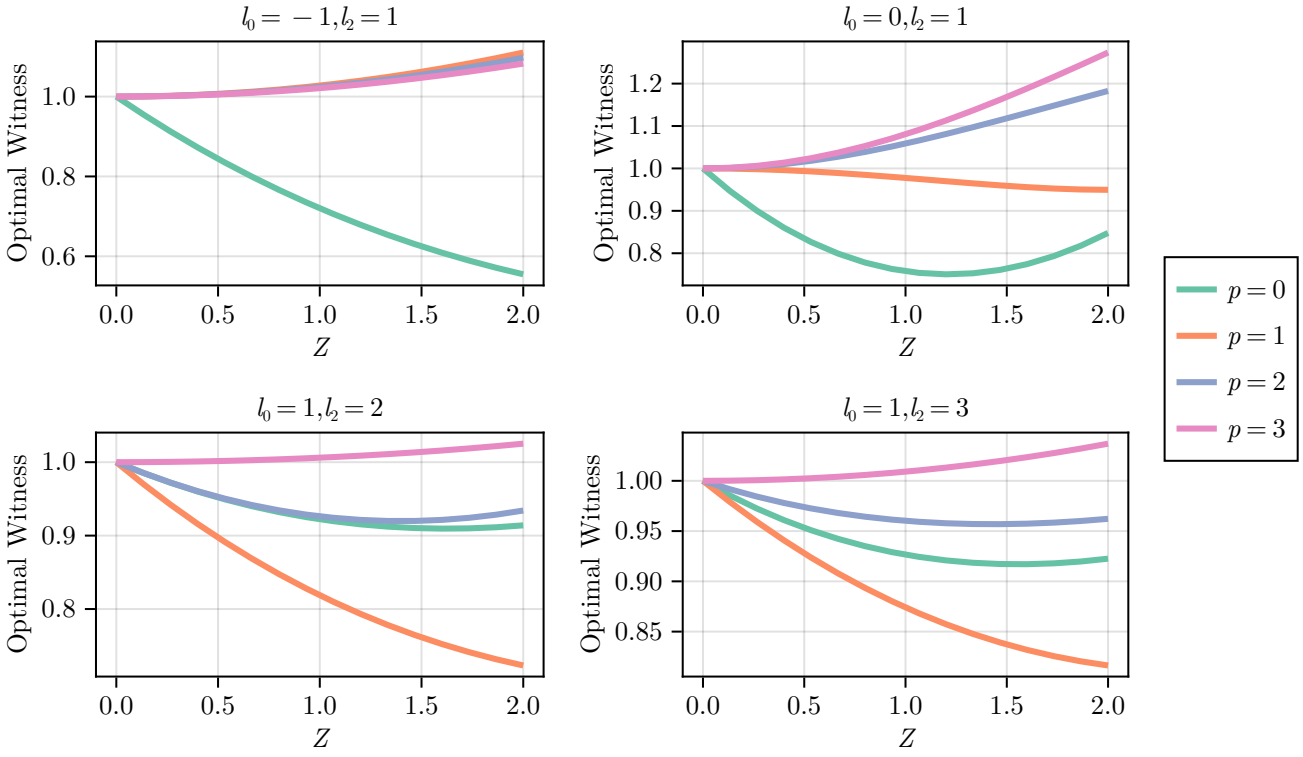


FIG. 2. Optimized witness value in the bipartite case.

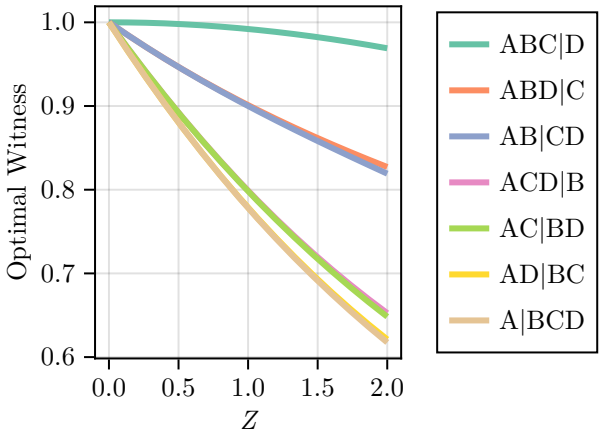


FIG. 3. Optimized witness value for bipartite entanglement over multiple modes with $l_0 = 2$ and $l_2 = 1$, so that $l_1 = 2l_0 - l_2 = 3$ and $p_{max} = \min(2l_0, l_2) = 1$.

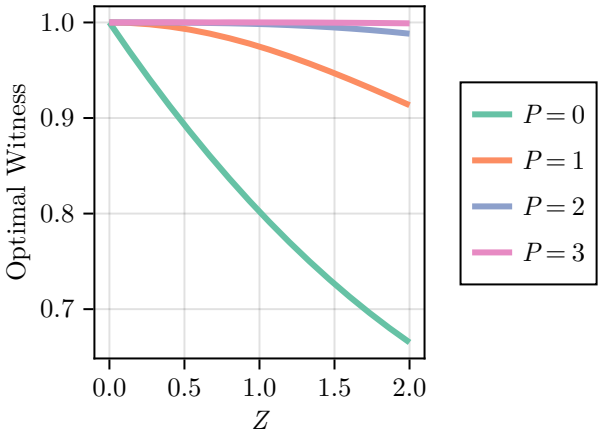


FIG. 4. Optimized witness value for genuine multipartite entanglement with $l_0 = 2$ and $l_2 = 1$, so that $l_1 = 2l_0 - l_2 = 3$ and $p_{max} = \min(2l_0, l_2) = 1$.

beams. The evolution of the mode structure in the nonlinear interaction is governed by chiral selection rules, which dictate the appearance of radial modes with in the mean field. Moreover, we show that the radial structure of other OAM components can exhibit bi- and multipartite entanglement, depending on the relative chirality of the interacting modes. The first-order results are complemented by finite-distance calculations using the diffraction-free Bogoliubov solution, within the regime of negligible diffraction and small fluctuation backreaction. Our results can be investigated by current four-wave mixing and self-phase modulation setups in the softly focused

regime, where the Rayleigh distance is much larger than the nonlinear interaction distance. This effect can be useful for building new quantum information platforms, where bi- and multipartite entanglement are controlled by orbital angular momentum.

Appendix A: Diffraction-free Bogoliubov evolution

The first-order calculation in the main text retains diffraction and all nonlinear fluctuation terms in Eq. (25). Here we start from the same full fluctuation equation and adopt a different approximation: we neglect diffraction over the interaction region and discard the quadratic and cubic fluctuation terms R of Eq. (26), while retaining the full dependence on propagation distance. The resulting Bogoliubov equations are solved without expanding in the propagation distance or the accumulated classical nonlinear phase.

With the same decomposition $\mathcal{E} = \alpha_c + \eta$ and input $\alpha(\mathbf{r}) = \alpha_c(\mathbf{r}, 0)$, these equations are

$$\begin{aligned} 2ik\partial_z\alpha_c &= g|\alpha_c|^2\alpha_c, \\ 2ik\partial_z\eta &= g\left(2|\alpha_c|^2\eta + \alpha_c^2\eta^\dagger\right). \end{aligned} \tag{A1}$$

The classical intensity is constant along propagation. Defining

$$I(\mathbf{r}) = |\alpha(\mathbf{r})|^2, \qquad \theta(\mathbf{r}, z) = \frac{gz}{2k}I(\mathbf{r}), \tag{A2}$$

we have

$$\alpha_c(\mathbf{r}, z) = \alpha(\mathbf{r})e^{-i\theta(\mathbf{r},z)}. \tag{A3}$$

1. Local fluctuation solution

Each transverse position evolves independently. Introduce the rotating-frame fluctuation $b(\mathbf{r}, z) = e^{i\theta(\mathbf{r},z)}\eta(\mathbf{r}, z)$. Suppressing the position argument temporarily, Eq. (A1) becomes

$$2ik\partial_z b = g\left(Ib + \alpha^2 b^\dagger\right). \tag{A4}$$

Together with its conjugate, this gives

$$2ik\partial_z \begin{pmatrix} b \\ b^\dagger \end{pmatrix} = gM \begin{pmatrix} b \\ b^\dagger \end{pmatrix}, \qquad M = \begin{pmatrix} I & \alpha^2 \\ -\alpha^{*2} & -I \end{pmatrix}. \tag{A5}$$

The matrix is independent of z and satisfies $M^2 = 0$. Therefore its exponential terminates exactly:

$$\exp\left(-\frac{igz}{2k}M\right) = \mathbb{I} - \frac{igz}{2k}M. \tag{A6}$$

Returning to η and writing $\eta_0(\mathbf{r}) = \eta(\mathbf{r}, 0)$, we obtain

$$\begin{aligned} \eta(\mathbf{r}, z) = e^{-i\theta}\Big[&(1 - i\theta)\eta_0(\mathbf{r}) \\ &- i\frac{gz}{2k}\alpha^2(\mathbf{r})\eta_0^\dagger(\mathbf{r})\Big]. \end{aligned} \tag{A7}$$

Although the rotating-frame coefficients are linear in z, this is not a first-order Taylor approximation. It is the exact solution of the diffraction-free linearized fluctuation equations.

Writing $\eta = A\eta_0 + B\eta_0^\dagger$, the coefficients are

$$A = e^{-i\theta}(1 - i\theta), \qquad B = -i\frac{gz}{2k}\alpha^2 e^{-i\theta}. \tag{A8}$$

They obey $|A|^2 - |B|^2 = 1$, so the field commutator is preserved. For the coherent input, $\langle\eta_0\rangle = 0$ and hence $\langle\eta(z)\rangle = 0$. Thus the mean field equals α_c within this approximation.

2. Field and mode correlations

The initial fluctuations satisfy

$$\begin{aligned} \langle\eta_0(\mathbf{r}_1)\eta_0^\dagger(\mathbf{r}_2)\rangle &= \delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2), \\ \langle\eta_0^\dagger(\mathbf{r}_2)\eta_0(\mathbf{r}_1)\rangle &= \langle\eta_0(\mathbf{r}_1)\eta_0(\mathbf{r}_2)\rangle = 0. \end{aligned} \tag{A9}$$

Consequently the correlations are

$$\begin{aligned} F(\mathbf{r}_1, \mathbf{r}_2; z) &= A(\mathbf{r}_1)B(\mathbf{r}_2)\delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2), \\ H(\mathbf{r}_1, \mathbf{r}_2; z) &= B(\mathbf{r}_1)B^*(\mathbf{r}_2)\delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2). \end{aligned} \tag{A10}$$

Substituting the coefficients gives

$$\begin{aligned} F(\mathbf{r}_1, \mathbf{r}_2; z) = &\left[-\frac{igz}{2k} - \left(\frac{gz}{2k}\right)^2 I(\mathbf{r}_1)\right] \\ &\times \alpha_c^2(\mathbf{r}_1, z)\delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2), \\ H(\mathbf{r}_1, \mathbf{r}_2; z) = &\left(\frac{gz}{2k}\right)^2 I^2(\mathbf{r}_1)\delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2). \end{aligned} \tag{A11}$$

Using the main-text scaling, $\theta = Z|u|^2$ and $u_Z = u\exp(-iZ|u|^2)$, projection onto the detection modes yields

$$\begin{aligned} f(v_1, v_2) = &- iZ(v_1, v_2; u_Z, u_Z) \\ &- Z^2(v_1, v_2, u_Z; u_Z, u_Z, u_Z), \\ h(v_1, v_2) = &\, Z^2(v_1, u_Z, u_Z; v_2, u_Z, u_Z). \end{aligned} \tag{A12}$$

The phases in h cancel, whereas those in f retain the nonlinear classical evolution. Expanding Eq. (A12) at small Z recovers $f = -iZ(v_1, v_2; u, u) + O(Z^2)$ and $h = O(Z^2)$, in agreement with the first-order calculation in the main text. At finite distance, these expressions are exact only within the diffraction-free Bogoliubov approximation, not for the full quantum Kerr equation.

The approximation requires negligible diffraction of the relevant transverse structures over the interaction length, as well as sufficiently small fluctuation interactions and backreaction on the classical field. A large pump Rayleigh range is a useful initial scale, but finer structure generated by the nonlinear phase can have shorter diffraction lengths. The derivation applies to any

coherent input profile, including a coherent pump plus seeds.

The Bogoliubov approximation also permits retaining diffraction in both the classical background and the linear fluctuation equations. Quantum correlations can then be computed numerically by propagating the corresponding Green functions [42, 43], or by back-propagating the detection functions with the adjoint fluctuation equations [44, 45] and evaluating their correlations in the known input state. Such a numerical treatment with diffraction lies beyond the scope of the present work, which uses the analytical diffraction-free solution above.

Appendix B: Coherent seeding and first-order selection rules

1. Role of the seed in quadrature detection

For the single coherent pump $\alpha = A_p v_p$, with $v_p = LG_{0,l_p}(w)$, both terms in Eq. (B3) with $\epsilon = 0$ vanish for a detection mode of definite OAM $l_v \neq l_p$: the nonlinear mean-field source $|v_p|^2 v_p$ carries the same OAM as the pump. Thus, within this first-order treatment, self-phase modulation changes the mean field only in the $l = l_p$ sector, including its radial structure.

The two-mode correlations studied in the main text instead involve $l_1 + l_2 = 2l_p$ with $l_2 \neq l_p$, so neither detected mode has a coherent mean at this order. This does not prevent quadrature measurements with an external local oscillator, but a coherent amplitude in a detected mode can assist spatial alignment and phase control. The purpose of the seed is to supply such a reference while retaining, as closely as possible, the original entanglement correlations. Zero coherent mean should not be identified with exactly zero photon occupation: the centered population $h(v, v)$ starts beyond first order in distance.

We consider mutually coherent, collinear pump and seed fields sharing the frequency, polarization, and temporal mode implicit in our scalar model. We first derive the first-order correlations for an arbitrary spatial seed, and then examine seeding only v_2 or directly seeding both detected OAM families. Throughout this appendix, "first order" in propagation refers to the coefficient linear in distance; terms linear and quadratic in seed amplitude are distinguished explicitly.

2. First-order corrections for an arbitrary coherent seed

Let the input profile be

$$\alpha(\mathbf{r}, 0) = A_p\big[v_p(\mathbf{r}) + \epsilon s(\mathbf{r})\big], \tag{B1}$$

where $\|v_p\| = \|s\| = 1$ and the complex parameter ϵ specifies the overall seed amplitude and phase relative to the pump. The normalized profile s specifies its spatial composition; for example, $s = \sum_j c_j v_{s,j}$ in an orthonormal seed basis, with $\sum_j |c_j|^2 = 1$. We use the pump waist and occupation $N_p = A_p^2$ to define

$$\begin{aligned} U_p &= w v_p, \\ S &= w s, \\ U &= U_p + \epsilon S, \\ Z_p &= \frac{g z N_p}{2kw^2}. \end{aligned} \tag{B2}$$

The profiles U_p, S, U are dimensionless, and Z_p is held fixed when comparing different seeds. Weak seeding means $|\epsilon| \ll 1$; it need not imply a seed occupation smaller than one. If the seed is orthogonal to the pump, its occupation relative to the pump is $q = |\epsilon|^2$, and the main-text normalization is $u = U/\sqrt{1+q}$, $Z = (1+q)Z_p$. More generally, replace $1 + q$ by $\|v_p + \epsilon s\|^2$. These conventions give the same product $Zu^2 = Z_p U^2$.

The input remains a coherent state of a spatial field distribution. Therefore, the mean-field formula and the general correlation derivation apply directly to the combined profile. For example, in freely propagated detection modes, with the nonlinear overlap evaluated at the input plane,

$$\langle a(v, z)\rangle = \frac{A_p}{w}\left[(v, U) - iZ_p(v, U; U, U)\right] + O(Z_p^2). \tag{B3}$$

For orthonormal detection modes v_1, v_2, define

$$\begin{aligned} C_{12} &= (v_1, v_2; U_p, U_p), \\ \delta C_{12}^{(1)} &= 2\epsilon(v_1, v_2; U_p, S), \\ \delta C_{12}^{(2)} &= \epsilon^2(v_1, v_2; S, S). \end{aligned} \tag{B4}$$

The superscripts indicate order in seed amplitude, not distance. Expanding $U^2 = (U_p + \epsilon S)^2$ gives

$$\begin{aligned} f(v_1, v_2) &= -iZ_p\left[C_{12} + \delta C_{12}^{(1)} + \delta C_{12}^{(2)}\right] + O(Z_p^2), \\ h(v_1, v_2) &= O(Z_p^2), \\ D_{\text{opt}} &= 1 - 2|Z_p|\left|C_{12} + \delta C_{12}^{(1)} + \delta C_{12}^{(2)}\right| + O(Z_p^2). \end{aligned} \tag{B5}$$

This is the complete coefficient linear in distance for an arbitrary coherent seed. It contains no terms beyond quadratic order in seed amplitude. Because f and h are centered correlations, the seed's coherent displacement alone does not contribute to them; the corrections above arise from its participation in the nonlinear interaction.

The angular content determines which corrections survive. Write

$$U_p = R_p(r)e^{il_p\phi}, \qquad S = \sum_m S_m(r)e^{im\phi}. \tag{B6}$$

For detection OAM sum $L = l_1 + l_2$, the pump–seed term requires $l_p + m = L$, and the seed–seed term requires $m + n = L$. In the original channel $L = 2l_p$, the relevant

harmonic is

$$[U^2]_{2l_p} = e^{2il_p\phi}\left[R_p^2 + 2\epsilon R_p S_{l_p} + \epsilon^2 \sum_m S_m S_{2l_p-m}\right]. \tag{B7}$$

Thus only a seed component with pump OAM can modify this channel linearly in seed amplitude. Quadratic corrections require complementary seed OAM values summing to $2l_p$. The radial overlaps must also be nonzero. In particular, a seed with pump OAM but a different radial profile can alter the radial rule at linear order in seed amplitude; a seed proportional to v_p simply changes the combined pump amplitude.

3. Seeding only the mode v_2

Consider the same detection family as in the main text,

$$\begin{aligned} v_p &= LG_{0,l_p}(w), \\ v_2 &= LG_{0,l_2}(w_2), \\ v_{1,p} &= LG_{p,l_1}(w_1), \\ l_1 &= 2l_p - l_2, \\ \frac{1}{w_1^2} &= \frac{2}{w^2} + \frac{1}{w_2^2}, \qquad l_2 \neq l_p. \end{aligned} \tag{B8}$$

Here l_p denotes pump OAM, whereas the index p of $v_{1,p}$ denotes radial order. Set $s = v_2$, so $S = U_2$, where $U_2 = wv_2$. The pump–seed and seed–seed terms carry OAM $l_p + l_2$ and $2l_2$, respectively. Neither equals $2l_p$, so both corrections vanish in the original channel. Defining $C_p = (v_{1,p}, v_2; U_p, U_p)$ gives

$$\begin{aligned} f(v_{1,p}, v_2) &= -iZ_pC_p + O(Z_p^2), \\ D_{\text{opt}} &= 1 - 2|Z_pC_p| + O(Z_p^2). \end{aligned} \tag{B9}$$

The coefficient linear in distance is independent of ϵ, including its phase. Hence the original cutoff

$$p_{\max} = |l_p| + \frac{|l_2| - |2l_p - l_2|}{2} \tag{B10}$$

is preserved exactly at this order for the detection profiles in (B8): $C_p = 0$ for $p > p_{\max}$. The seed does open other OAM channels, at orders $Z_p\epsilon$ and $Z_p\epsilon^2$, but these do not enter the selected pair's first-order coefficient.

The mean-field benefit is twofold. The seeded mode has the input amplitude

$$\langle a(v_2, 0)\rangle = A_p\epsilon. \tag{B11}$$

Moreover, the nonlinear source $|U|^2U$ contains $\epsilon^* U_p^2 U_2^*$, whose OAM is $2l_p - l_2 = l_1$. Projection of (B3) therefore yields

$$\langle a(v_{1,p}, z)\rangle = -iZ_pA_p\epsilon^*C_p + O(Z_p^2). \tag{B12}$$

A seed in v_2 can thus generate coherent references in its conjugate radial modes while preserving the original first-order correlations. These stimulated amplitudes are suppressed by Z_p, follow the radial coefficients C_p, and vanish at this order outside the cutoff. Whether they suffice for homodyne phase control depends on the experimental sensitivity.

4. Seeding both detected OAM families

To supply a direct coherent reference in every selected detection mode, including radial modes whose stimulated amplitudes are too small, take

$$S = c_2U_2 + \sum_{p\in\mathcal{S}} c_{1,p}U_{1,p}, \tag{B13}$$

where $U_{1,p} = wv_{1,p}$ and $\mathcal{S}$ is the finite set of seeded radial orders. The two OAM families and the pump are mutually orthogonal, as are the radial modes within the first family. Normalization of s requires

$$|c_2|^2 + \sum_{p\in\mathcal{S}} |c_{1,p}|^2 = 1, \tag{B14}$$

the seed power fraction is $q = |\epsilon|^2$, and the input means are $\langle a(v_2, 0)\rangle = A_p\epsilon c_2$ and $\langle a(v_{1,p}, 0)\rangle = A_p\epsilon c_{1,p}$. These directly injected amplitudes are not suppressed by propagation distance.

Neither seed family has pump OAM, so $\delta C^{(1)}$ still vanishes in the original channel. However, their OAM values are complementary: $l_1 + l_2 = 2l_p$. Equation (B7) now gives

$$[U^2]_{2l_p} = U_p^2 + 2\epsilon^2 c_2 \sum_{p\in\mathcal{S}} c_{1,p}U_2U_{1,p}. \tag{B15}$$

For a detected radial order r, define

$$\begin{aligned} K_{rp} &= (v_{1,r}, v_2; U_2, U_{1,p}) \\ &= w^2 \int d^2\mathbf{r}\, v_{1,r}^* v_{1,p}|v_2|^2, \\ \delta C_r &= 2\epsilon^2 c_2 \sum_{p\in\mathcal{S}} c_{1,p}K_{rp}. \end{aligned} \tag{B16}$$

The first-order correlations become

$$\begin{aligned} f(v_{1,r}, v_2) &= -iZ_p(C_r + \delta C_r) + O(Z_p^2), \\ D_{\text{opt}} &= 1 - 2|Z_p|\,|C_r + \delta C_r| + O(Z_p^2). \end{aligned} \tag{B17}$$

The correction is quadratic in seed amplitudes but remains linear in distance. Its phase relative to C_r depends on the products $\epsilon^2 c_2 c_{1,p}$, so complementary seeds can enhance or reduce the magnitude of an allowed correlation.

The weighted overlaps K_{rp} need not obey the original radial cutoff. Orthogonality of radial LG modes uses an unweighted inner product, whereas the factor $|v_2|^2$ changes the radial weight and Gaussian envelope. In particular,

$$K_{rr} = w^2 \int d^2\mathbf{r}\,|v_{1,r}|^2|v_2|^2 > 0 \tag{B18}$$

for overlapping LG profiles. Thus directly seeding $v_{1,r}$ and v_2 can remove a first-order zero even when $r > p_{\max}$ and $C_r = 0$. Multiple radial seeds may interfere, with cancellation possible only for particular amplitudes and phases.

For a fixed normalized seed profile, $\delta C_r = O(\epsilon^2)$ and $q = |\epsilon|^2$. The original radial rule is therefore preserved through linear order in seed amplitude, but generally modified at quadratic order in seed amplitude, or first order in seed power fraction at fixed seed composition. For an allowed pair, the condition

$$|\delta C_r| \ll |C_r| \tag{B19}$$

ensures that its first-order correlation is approximately unchanged. For a forbidden pair, any nonzero δC_r lifts the first-order zero, however small its absolute magnitude.

Seeding only v_2 therefore preserves the original first-order radial rule exactly while relying on stimulated means in the complementary family. Seeding both families supplies independent input references in all selected modes, at the cost of quadratic seed corrections to that rule. These statements concern the first-order two-mode criterion; they do not establish a multipartite selection rule or a finite-distance witness violation. Beyond the small-distance regime, the combined input may be inserted directly into the correlation formulas of Appendix A, within their stated assumptions.

Appendix C: Working note: pairwise and collective-mode witnesses for multimode bipartitions

Purpose and scope. This appendix is a detailed working argument for consideration, not proposed final manuscript text. It first shows that an individual entangled pair crossing a partition already proves entanglement of that partition, then constructs a collective-mode Duan witness as a quantitative refinement. Both statements are related to the radial selection rule. For the four-mode example in the main text, it proves small-distance entanglement across six bipartitions and explains why the seventh cannot have a linear witness violation. It does not prove the quadratic violation for the seventh partition, identify the numerically optimal witnesses, or by itself establish genuine multipartite entanglement of a mixed reduced state. Appendix D adds a genuine multipartite conclusion for a pure global state with the unobserved modes assigned to one party.

1. Input assumptions and first-order correlations

Consider a finite set of orthonormal detection modes, with the coherent means subtracted when defining f and h. The first-order calculation in the main text, Eq. (32), gives

$$f(v_i, v_j) = -iZC_{ij} + O(Z^2), \qquad h(v_i, v_j) = O(Z^2), \tag{C1}$$

where

$$C_{ij} = (v_i, v_j; u, u) = \int d^2\mathbf{r}\, v_i^* v_j^* u^2. \tag{C2}$$

The detection profiles and input shape are fixed as $Z \to 0$. The dimensionless variable Z is proportional to z at fixed input and medium parameters. No finite-distance Bogoliubov approximation is required for the leading coefficient in Eq. (C1): it follows from the first-order displaced quantum calculation for a coherent input. As elsewhere, continuum expressions are interpreted through finite, smooth mode projections within the effective model.

For the unseeded example, the selected modes are

$$\begin{aligned} v_A &= LG_{0,1}(w), \quad & v_B &= LG_{0,3}(w_1), \\ v_C &= LG_{1,3}(w_1), \quad & v_D &= LG_{2,3}(w_1), \end{aligned} \tag{C3}$$

with pump $LG_{0,2}(w)$, $w_1 = w/\sqrt{3}$, and $p_{\max} = 1$. Define

$$c_B = C_{AB}, \qquad c_C = C_{AC}. \tag{C4}$$

The angular selection rule eliminates all entries among B, C, D, as well as C_{AA}; the radial rule gives $C_{AD} = 0$. In the ordering A, B, C, D,

$$C = \begin{pmatrix} 0 & c_B & c_C & 0 \\ c_B & 0 & 0 & 0 \\ c_C & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{C5}$$

This statement concerns the first-order anomalous correlations of the selected modes. It does not assert that these modes form a closed subsystem of the full field.

2. Simpler existence proof using an individual pair

A collective mode is not needed to prove entanglement across a multimode bipartition. Fix a cut $S|T$ with $A \in S$, and suppose that one original detection mode $v_j \in T$ has $c_j = C_{Aj} \neq 0$. The ordinary pairwise criterion gives

$$\begin{aligned} D_{A,j} &= 1 + h(v_A, v_A) + h(v_j, v_j) - 2|f(v_A, v_j)| \\ &= 1 - 2|Z|\,|c_j| + O(Z^2). \end{aligned} \tag{C6}$$

It is therefore below one for sufficiently small nonzero Z. This is a linear violation with a second-order remainder, not a calculation of the violation through second order.

If the full selected state were separable across $S|T$, tracing out all modes except A, j would preserve separability across $A|j$. The observed pairwise violation contradicts this. Hence one entangled pair crossing the cut suffices to certify that cut. Neither purity of the selected state nor a change of detection basis is required.

In the four-mode example, the AB pair certifies $A|BCD$, $AD|BC$, $ACD|B$ and $AC|BD$. The AC pair additionally certifies $ABD|C$ and $AB|CD$. Neither crosses $ABC|D$, so that cut remains undecided by the first-order pairwise argument.

More generally, if a selected mode A is entangled with every mode $B_1, \ldots, B_N$, every nontrivial bipartition of this selected set places at least one B_j opposite A. The selected state is therefore entangled across every bipartition, a property called full inseparability. For a mixed reduced state this alone does not exclude mixtures of states separable across different cuts. Appendix D explains how global purity establishes a genuine multipartite statement when the unobserved modes are included in one of the parties.

The following collective-mode construction is an optional quantitative refinement. Among individual pairs crossing a cut, the strongest first-order Duan coefficient is

$$G_{\mathrm{pair}} = \max_{j\in R} |c_j|, \tag{C7}$$

where R contains the opposite-side partners. Combining these correlations gives

$$G_R = \sqrt{\sum_{j\in R} |c_j|^2} \geq G_{\mathrm{pair}}. \tag{C8}$$

The two constructions certify the same existence of small-distance entanglement whenever at least one coefficient is nonzero; the collective one gives a larger leading violation when more than one coefficient contributes. It is useful for quantitative comparison with witness curves, but is not logically necessary for the partition argument.

3. A collective mode on the opposite side of a partition

Fix a bipartition $S|T$, and label the sides so that $A \in S$. Let R be the set of modes among B, C that lie in T. Set

$$G_R = \left(\sum_{j\in R} |c_j|^2 \right)^{1/2}. \tag{C9}$$

If $G_R > 0$, define

$$v_R = \frac{1}{G_R} \sum_{j\in R} c_j v_j. \tag{C10}$$

Orthogonality of the original modes gives $(v_R; v_R) = 1$ and $(v_A; v_R) = 0$. The collective mode is supported entirely on party T. It can be completed to an orthonormal mode basis within that party, so replacing the original modes of T by this basis is a local passive transformation relative to $S|T$.

The complex conjugations matter. The annihilation operator is antilinear in its profile:

$$a(v_R) = \frac{1}{G_R} \sum_{j\in R} c_j^* a(v_j). \tag{C11}$$

Therefore

$$\begin{aligned} f(v_A, v_R) &= \frac{1}{G_R} \sum_{j\in R} c_j^* f(v_A, v_j) \\ &= -iZG_R + O(Z^2). \end{aligned} \tag{C12}$$

The phases of the profile coefficients have aligned the pair correlations constructively. Meanwhile,

$$h(v_R, v_R) = \frac{1}{G_R^2} \sum_{j,k\in R} c_j^* c_k h(v_j, v_k) = O(Z^2). \tag{C13}$$

No assumption about the purity of the selected two-mode state is needed.

Among normalized superpositions on these opposite-side modes, this choice maximizes the magnitude of the first-order correlation with A. Indeed, for $v_d = \sum_{j\in R} d_j v_j$ and $\sum_j |d_j|^2 = 1$, Cauchy–Schwarz gives

$$\left| \sum_{j\in R} d_j^* c_j \right| \leq G_R, \tag{C14}$$

with equality for the coefficients in Eq. (C10), up to a common phase. This is optimality within this collective-mode construction, not a claim of optimality over all covariance witnesses.

4. An explicit entanglement certificate

For two orthogonal normalized modes, the phase-optimized Duan expression is

$$\begin{aligned} D_{A,R} &= 1 + h(v_A, v_A) + h(v_R, v_R) - 2|f(v_A, v_R)| \\ &= 1 - 2|Z|G_R + O(Z^2). \end{aligned} \tag{C15}$$

The first equality is exact for the second moments of the state, including any correlations with unobserved modes. Self-squeezing and normal cross-correlations cancel in this particular sum of quadrature variances; their absence does not require additional assumptions about the state.

For a fixed input and $G_R > 0$, the nonzero linear decrease dominates the $O(Z^2)$ remainder at sufficiently small nonzero Z. Hence $D_{A,R} < 1$ in a neighborhood of the input plane. This proves entanglement of the reduced pair A, v_R.

It also proves entanglement across the original multimode partition. If the selected state were separable across $S|T$, a local change of mode basis within T followed by discarding every mode except A and v_R would

leave a separable two-mode state. Such a state cannot violate the Duan bound. Thus the constructed violation contradicts separability across $S|T$, regardless of correlations within either party or mixedness caused by tracing out other spatial modes.

This is a constructive statement: it supplies a witness and proves its violation. A nonzero cross-correlation alone would not be an entanglement proof. Here the leading anomalous correlation is linear in Z, whereas the normal populations that enter the same criterion begin at Z^2.

5. The witness matrix and its relation to the partition

For the phase-adjusted pair, order the quadratures as (X_A, P_A, X_R, P_R) and use the manuscript convention $\gamma_{jk} = \langle\{\Delta r_j, \Delta r_k\}\rangle$, with vacuum covariance $\mathbb{I}$. The Duan matrix is

$$W_D = \frac{1}{4}\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}. \tag{C16}$$

A local phase of v_A can be chosen to make $f(v_A, v_R)$ real and negative. At leading order, a fixed phase suffices for either fixed sign of Z; choosing the phase from the full f gives the exact phase optimization in Eq. (C15).

Let L be the real matrix extracting these four quadratures from the full selected-mode quadrature vector after the local basis and phase changes. Then

$$\gamma_{A,R} = L\gamma L^T, \qquad \mathcal{Z}_{S|T} = L^T W_D L, \tag{C17}$$

and

$$D_{A,R} = \mathrm{Tr}(\mathcal{Z}_{S|T}\gamma). \tag{C18}$$

For an explicit construction, a profile coefficient d_j in $v = \sum_j d_j v_j$ contributes the block

$$\begin{pmatrix} \mathrm{Re}\, d_j & \mathrm{Im}\, d_j \\ -\mathrm{Im}\, d_j & \mathrm{Re}\, d_j \end{pmatrix} \tag{C19}$$

to the rows extracting $(X(v), P(v))$. This fixes the phase convention and allows the analytical witness to be evaluated on the same covariance matrices as the numerical optimization.

The matrix $\mathcal{Z}_{S|T}$ is positive semidefinite and has trace one, since the extracted modes are orthonormal and $LL^T = \mathbb{I}_4$. It generally has off-diagonal blocks across the partition: those blocks measure the correlations responsible for the violation. It may also have entries between original modes belonging to the same party, because v_R is a superposition of those modes.

If w_{opt} denotes minimization over the full valid witness class for the same partition and normalization, then

$$w_{\mathrm{opt}}(Z) \le D_{A,R}(Z). \tag{C20}$$

This provides an explicit achievable witness value. Equality, including equality of the leading slopes, would require a separate optimality argument. In particular, the constructed witness should not be identified with the Fully-Wit output without comparison.

6. All seven bipartitions of the four-mode example

Assuming c_B and c_C are nonzero, the construction gives

Partition	G_R
$A\|BCD$	$\sqrt{\|c_B\|^2 + \|c_C\|^2}$
$AD\|BC$	$\sqrt{\|c_B\|^2 + \|c_C\|^2}$
$ACD\|B$	$\|c_B\|$
$AC\|BD$	$\|c_B\|$
$ABD\|C$	$\|c_C\|$
$AB\|CD$	$\|c_C\|$
$ABC\|D$	0

The first six rows have $D_{A,R} = 1 - 2|Z|G_R + O(Z^2)$ and thus certify entanglement at sufficiently small nonzero distance. A row with $G_R = 0$ does not define a collective mode through Eq. (C10); it records that this first-order construction supplies no violation.

The first four partitions in the strongest numerical group therefore have two distinct constructive slopes, not one. If $|c_B|$ is appreciably larger than $|c_C|$, the strengths $\sqrt{|c_B|^2 + |c_C|^2}$ and $|c_B|$ are close. The next two partitions use only the c_C coupling across their cut. This gives a quantitative explanation to test against the numerical curves, rather than assuming exact degeneracy within each visually identified group.

The last partition places every first-order coupled mode on the same side. Its absence of a linear violation can be established for the full normalized witness class, as follows.

7. Why the excluded-mode cut has no linear violation

For $S = ABC$ and $T = D$, write the exact physical selected covariance as

$$\gamma(Z) = \begin{pmatrix} \gamma_S(Z) & K(Z) \\ K(Z)^T & \gamma_D(Z) \end{pmatrix}. \tag{C21}$$

Both anomalous and normal cross-correlations vanish at first order, so $K(Z) = O(Z^2)$. The matrix

$$\gamma_{\mathrm{prod}}(Z) = \gamma_S(Z) \oplus \gamma_D(Z) \tag{C22}$$

is a physical covariance matrix of the product of the two actual marginal states. It is therefore separable across this cut, and every valid witness with threshold one satisfies

$$\mathrm{Tr}(\mathcal{Z}\gamma_{\mathrm{prod}}) \ge 1. \tag{C23}$$

For a fixed finite witness,

$$\begin{aligned}\mathrm{Tr}(\mathcal{Z}\gamma) &= \mathrm{Tr}(\mathcal{Z}\gamma_{\mathrm{prod}}) + \mathrm{Tr}[\mathcal{Z}(\gamma-\gamma_{\mathrm{prod}})] \\ &\geq 1 - O(Z^2).\end{aligned} \tag{C24}$$

Internal $O(Z)$ correlations within ABC do not evade this bound: they are already included in the product reference covariance.

One must also consider whether an optimized witness depending on Z could diverge and turn the small cross-block into a linear violation. Near the coherent input, $\gamma(0) = \mathbb{I}$, so $\gamma(Z) \geq \frac{1}{2}\mathbb{I}$ for sufficiently small $|Z|$. For any positive semidefinite candidate witness that achieves $\mathrm{Tr}(\mathcal{Z}\gamma) \leq 1$,

$$\tfrac{1}{2}\,\mathrm{Tr}\,\mathcal{Z} \leq \mathrm{Tr}(\mathcal{Z}\gamma) \leq 1. \tag{C25}$$

Hence $\mathrm{Tr}\,\mathcal{Z} \leq 2$, uniformly in this neighborhood. The bound

$$|\,\mathrm{Tr}(\mathcal{Z}\Delta\gamma)| \leq \mathrm{Tr}(\mathcal{Z})\,\|\Delta\gamma\|_{\mathrm{op}} \tag{C26}$$

then establishes the same $O(Z^2)$ limit for any violating optimized witness. Candidates with value above one do not certify entanglement. Thus a linear onset of optimized violation is excluded as well.

This result does not prove separability of $ABC|D$ at small nonzero Z. It only shows that any violation must begin at quadratic or higher order. It applies to the physical covariance family; treating its first-order truncation as an exact covariance matrix could violate the uncertainty relation and give misleading optimization results.

8. What is needed to establish the quadratic violation?

Within the diffraction-free Bogoliubov model, the coefficients required for a second-order analysis follow from Eq. (A12). Define

$$\begin{aligned}Q_{ij} &= \int d^2\mathbf{r}\, v_i^* v_j^* |u|^2 u^2, \\ R_{ij} &= \int d^2\mathbf{r}\, v_i^* v_j |u|^4.\end{aligned} \tag{C27}$$

Using $u_Z^2 = u^2\exp\!\big(-2iZ|u|^2\big)$ gives

$$\begin{aligned}f(v_i,v_j) &= -iZC_{ij} - 3Z^2Q_{ij} + O(Z^3), \\ h(v_i,v_j) &= Z^2R_{ij}.\end{aligned} \tag{C28}$$

The -3 includes -2 from expanding the phase in the linear term and -1 from the explicit quadratic term. The expression for h is exact within this diffraction-free linearized model, because its background phases cancel.

For A, D, the leading coefficient C_{AD} vanishes, but the additional radial weight in Q_{AD} need not obey the original radial cutoff. The ordinary two-mode criterion would give

$$D_{A,D} = 1 + Z^2\,(R_{AA} + R_{DD} - 6|Q_{AD}|) + O(|Z|^3). \tag{C29}$$

A negative quadratic coefficient would already certify $ABC|D$ through the reduced pair. A nonnegative coefficient is inconclusive: correlations involving B, C may allow another witness to detect entanglement of ABC with D even when this pairwise test fails.

A full analysis of the seventh curve therefore requires the second-order covariance and either a suitable collective witness or a partial-transpose analysis. Normal correlations and populations must be included at the same order as Q_{ij}. The present appendix does not establish the sign of that optimized second-order contribution. Moreover, these second-order coefficients belong to the diffraction-free Bogoliubov model; they are not asserted to be the full second-order coefficients of the untruncated quantum equation with diffraction.

9. Generalization and limits of the argument

For a selected mode v_A coupled to a larger radial family, the same proof uses

$$G_R = \left(\sum_{p\in R} |(v_A, v_p; u, u)|^2\right)^{1/2}. \tag{C30}$$

Every partition placing at least one nonzero partner coefficient opposite A admits the constructed first-order certificate. The OAM-dependent radial rule controls which coefficients can enter this sum, while their magnitudes determine the constructive witness slope. Modes excluded at first order can still acquire correlations and entanglement at higher orders. This statement does not require isolation from other spatial modes.

More generally, for any cross-partition coefficient matrix C_{ST}, choose normalized collective profiles with coefficient vectors x on S and y on T. Their leading anomalous correlation is $-iZx^\dagger C_{ST}y^*$. Maximizing its magnitude over the two normalized vectors gives the largest singular value $\sigma_{\max}(C_{ST})$. The same Duan construction yields

$$D_{S,T} = 1 - 2|Z|\sigma_{\max}(C_{ST}) + O(Z^2). \tag{C31}$$

For the star matrix (C5), the cross block has rank at most one, and its nonzero singular value is precisely G_R. This provides a general first-order certificate whenever the cross-partition pair-creation coefficient is nonzero, under the coherent-input assumptions of Eq. (C1).

The collective transformation must remain local relative to the partition being tested. Combining modes that lie on opposite sides would change the parties and would not prove entanglement of the original partition. Different cuts generally use different collective modes and different witnesses.

Finally, certifying all bipartitions with these separate witnesses is not a proof of genuine multipartite entanglement for a mixed selected state. A genuine multipartite witness must be valid for every biseparable decomposition simultaneously. A witness valid for that stronger task must, in particular, respect the $ABC|D$ bound, so it also cannot show a linear violation for the four-mode example. The first-order argument establishes the bipartite coupling structure and its witnesses. When the complete modeled field is pure, including the remaining modes in one party permits the genuine multipartite conclusion proved in Appendix D. The MultiWit certification after those modes are discarded is a separate, stronger statement about the reduced selected state.

Appendix D: Working note: genuine multipartite entanglement with the remaining modes grouped into one party

Purpose and scope. This working appendix develops an analytical consequence of the pairwise selection rules. If a mode A is entangled with each of N orthogonal modes B_j, and the entire modeled field is in a pure state, the field is genuinely multipartite entangled for an $N+1$-party grouping in which all remaining modes are assigned to one of those parties. This differs from genuine multipartite entanglement of the reduced state of the selected single modes. The latter can be certified by the numerical witnesses after the remaining modes are traced out. The discussion is for consideration and clarification, not final manuscript prose.

1. Selected modes, the complement, and the parties

Let the selected single-mode subsystems be

$$S = \{A, B_1, \ldots, B_N\}, \qquad N \geq 2, \tag{D1}$$

with mutually orthonormal mode profiles. Denote their orthogonal spatial-mode complement by $\mathcal{C}$. Here $\mathcal{C}$ means all remaining modes, including the pump and any other radial or angular modes; it is not the individual mode labeled C in the four-mode example of the main text.

For the modeled bosonic field, the mode decomposition gives the Fock-space factorization

$$\mathcal{H} = \mathcal{H}_A \otimes \bigotimes_{j=1}^{N} \mathcal{H}_{B_j} \otimes \mathcal{H}_{\mathcal{C}}. \tag{D2}$$

The complement is defined as an orthogonal subspace, so one does not inadvertently include overlapping detection profiles as independent parties. A complete basis within the complement can be chosen arbitrarily.

The proposed grouping has $N+1$ parties:

$$A \mid B_1 \mid \cdots \mid B_{N-1} \mid (B_N\mathcal{C}). \tag{D3}$$

The last party contains one selected radial mode and the entire complement. Grouping means assigning these subsystems to one party, not tracing them out or physically mixing their modes. In particular, every degree of freedom of the pure modeled field is still included.

We assume the global state is pure,

$$\rho_{S\mathcal{C}}(z) = |\Psi(z)\rangle\langle\Psi(z)|. \tag{D4}$$

This holds in an ideal closed, lossless unitary model with a pure coherent input and vacuum in the initially unoccupied modes. It also holds for the full fluctuation state evolved from vacuum by a lossless quadratic Bogoliubov Hamiltonian, within that approximation. It need not hold for an optical state after absorption, added medium noise or averaging over uncontrolled input parameters. Global purity is a separate premise, not something established by the pairwise witness values.

2. Pairwise premise and a common small-distance interval

Assume the reduced pair AB_j is entangled for each j. In the first-order treatment, a sufficient condition is

$$c_j = (v_A, v_{B_j}; u, u) \neq 0, \tag{D5}$$

since Eq. (C6) gives

$$D_{A,B_j}(Z) = 1 - 2|Z|\,|c_j| + O(Z^2) < 1 \tag{D6}$$

for sufficiently small nonzero Z. Each of these is a statement about the actual reduced pair of the global state; it does not require the other modes to be absent or uncorrelated.

For a fixed finite number of nonzero coefficients, there is a common interval in which all these violations hold simultaneously. For example, if the remainder for pair j has magnitude at most $M_j Z^2$ near the input, then $M_j|Z| < 2|c_j|$ suffices for that pair. Taking a sufficiently small interval common to the finite set gives all pairwise entanglements at the same distance. The asymptotic argument does not determine how large this interval is, and a very small coefficient can restrict it substantially.

The proof below only requires the pairwise premise. It remains valid at any distance where all the pairs are certified and the global state is pure, even if their entanglement is established using the full finite-distance covariance rather than a Taylor expansion.

3. Every bipartition of the grouped parties is entangled

Take any nontrivial bipartition $L|R$ of the parties in Eq. (D3). Choose the notation so that the party A lies in L. Since R is nonempty, it contains at least one selected mode B_j: either as its own party or inside the enlarged party $B_N\mathcal{C}$.

Suppose, for contradiction, that the global state were separable across this cut:

$$\rho_{LR} = \sum_{\lambda} p_\lambda \rho_L^{(\lambda)} \otimes \rho_R^{(\lambda)}. \tag{D7}$$

Tracing out all subsystems except A, B_j gives

$$\rho_{AB_j} = \sum_{\lambda} p_\lambda \rho_A^{(\lambda)} \otimes \rho_{B_j}^{(\lambda)}, \tag{D8}$$

which is separable. This contradicts the entanglement of that pair. Therefore the global state is entangled across every bipartition of the grouped parties.

This part of the proof does not use purity. It follows only from the fact that partial traces within each party preserve separability. If $B_j = B_N$, discarding $\mathcal{C}$ is one of these local partial traces, so correlations involving the complement do not invalidate the argument.

4. Where purity establishes genuine multipartite entanglement

For a mixed state, entanglement across every fixed bipartition is not sufficient to exclude a biseparable decomposition

$$\rho = \sum_{\lambda} p_\lambda \sigma_\lambda, \tag{D9}$$

where σ_λ is separable across a cut that may vary with λ. This is the distinction between full inseparability and genuine multipartite entanglement.

A pure state does not have this freedom. Every positive component in a convex decomposition of the rank-one density matrix $|\Psi\rangle\langle\Psi|$ must have support in the same one-dimensional subspace. Consequently, if a pure state were biseparable, that pure state itself would be separable across at least one fixed bipartition. Since the preceding argument excludes every such bipartition, the pure global state is genuinely multipartite entangled with respect to Eq. (D3).

The resulting sufficient statement is:

> For a pure state of the full modeled field, if A is entangled with each of N distinct modes B_j, then the state is genuinely multipartite entangled among the $N+1$ parties obtained by keeping $A, B_1, \ldots, B_{N-1}$ as individual parties and grouping B_N with all remaining modes.

Neither Gaussianity nor optimality of the pairwise witnesses is required for this logical implication. The small-distance expansion supplies one way to establish its pairwise premise. The theorem is applied to a physical pure-state family, not to a truncated density matrix that might fail positivity or normalization.

5. What the OAM-dependent radial rule establishes

In the unseeded configuration, A can be the selected mode $LG_{0,l_2}(w_2)$ and the partners the allowed modes $LG_{p,l_1}(w_1)$, with $l_1 = 2l_0 - l_2$ and the waist relation used in the main text. Let

$$N = \#\{p : (v_A, v_{B_p}; u, u) \neq 0\} \tag{D10}$$

count the nonzero coefficients in the selected radial family. If every coefficient from $p = 0$ through $p_{\max}$ is nonzero, then $N = p_{\max} + 1$ and the grouped theorem certifies $p_{\max} + 2$ parties. If an additional zero occurs, only the nonzero coefficients count toward this particular guarantee.

Thus the magnitudes and relative signs of the OAM values control how many resolved radial modes can serve as distinct parties in this first-order certification. The waist choice remains part of the statement. The result is a guaranteed grouping, not a claim that the entire field contains exactly this number of entangled modes or that other modes are separable. Higher-order couplings can involve further radial orders, and the proof places no upper bound on the total entanglement structure.

For the example $l_0 = 2$, $l_2 = 1$, the two nonzero partners are $B_1 = LG_{0,3}(w_1)$ and $B_2 = LG_{1,3}(w_1)$. Pairwise entanglement of A with both partners and global purity establish genuine tripartite entanglement of

$$A \mid B_1 \mid (B_2\mathcal{C}). \tag{D11}$$

The mode $LG_{2,3}(w_1)$, denoted D in the main text, belongs to $\mathcal{C}$ here, together with the pump and all other unselected modes. The absence of its first-order pairwise violation does not obstruct this three-party proof, because D has not been introduced as a separate party. Certifying a grouping in which D is a fourth distinct party would require an additional argument.

6. Freedom to assign the remaining modes

There is nothing special about placing the complement with B_N. It may instead be assigned to A or any other B_j. More generally, its orthogonal mode subspaces may be distributed among the parties, provided the grouping includes the whole pure system and keeps each of $A, B_1, \ldots, B_N$ in a distinct party. Every nontrivial cut then still separates A from at least one entangled partner, and the same proof applies.

This freedom must not be confused with certifying genuine entanglement of every individual mode in the complement. Some of those modes may be in vacuum or uncorrelated with the selected set. A party can contain unentangled ancillary modes without invalidating genuine entanglement among the grouped parties. Conversely, making the entire complement an additional standalone party would introduce the cut $S|\mathcal{C}$, whose entanglement is not established by the selected pairwise data.

7. Discarding the complement: the separate numerical claim

The reduced state measured on the selected single modes is

$$\rho_S = \mathrm{Tr}_{\mathcal{C}} |\Psi\rangle\langle\Psi|. \tag{D12}$$

It generally is mixed. The pairwise entanglements still prove that ρ_S is entangled across every bipartition, but the pure-state step above can no longer be applied. Grouping the complement and discarding it are different operations:

$$\begin{array}{ll} A|B_1|\cdots|(B_N\mathcal{C}) & \text{includes the full pure state,} \\ A|B_1|\cdots|B_N & \text{uses the reduced state } \rho_S. \end{array} \tag{D13}$$

Tracing out $\mathcal{C}$ is a local operation on the enlarged last party. Such an operation cannot create genuine multipartite entanglement, but it can destroy it. The grouped theorem therefore does not guarantee that the reduced state is genuinely multipartite entangled.

A valid MultiWit violation for the covariance of ρ_S supplies precisely the missing certification: it excludes all biseparable mixtures of the selected modes, even though their state is mixed. In parameter ranges where that violation is found, the genuine multipartite entanglement remains after the complement is discarded. This is a stronger accessible-mode statement than the grouped theorem. It also establishes the grouped state's genuine entanglement without needing to infer it from purity, since a local partial trace cannot create it.

The numerical result does not imply that $\mathcal{C}$ was uncorrelated, or that the reduced state became pure. It establishes robustness of the multipartite certification to discarding those modes. The analytical and numerical statements should be compared for the same selected set and at distances where their respective premises have been checked; the asymptotic pairwise proof alone does not establish the entire finite-distance interval of a numerical plot.

8. Limits and a possible role in the manuscript

Global purity refers to the complete system actually included in the claim. If the optical field is mixed because of losses or medium noise, placing the other optical modes into $\mathcal{C}$ does not restore purity. One could formally enlarge the system to include a purifying environment, but the last party would then include that environment, not just the spatial optical modes. That would be a different operational claim. The purely optical grouped theorem is therefore naturally a statement about the ideal lossless model; direct numerical witnesses of the reduced optical state remain the appropriate certification when additional noise is modeled.

For the ideal model, the argument supplies a short route from the first-order selection rule to a genuine multipartite statement. The individual pairwise Duan violations establish all bipartitions of the grouped system, and purity excludes biseparable mixtures. The optional collective-mode construction in Appendix C can improve the quantitative bipartite witness but is not needed for this proof.

A possible presentation would first establish the OAM-controlled pairwise radial support, then state the grouped pure-state consequence with its parties explicitly identified. The numerical multipartite section would ask the stronger question of whether the selected single modes remain genuinely multipartite entangled after all unselected modes are traced out. This organization connects the analytical selection rules to the numerical certification without claiming that pairwise entanglement alone proves genuine multipartite entanglement of an arbitrary mixed reduced state.

[1] A. E. Willner, H. Huang, Y. Yan, Y. Ren, N. Ahmed, G. Xie, C. Bao, L. Li, Y. Cao, Z. Zhao, J. Wang, M. P. J. Lavery, M. Tur, S. Ramachandran, A. F. Molisch, N. Ashrafi, and S. Ashrafi, Optical communications using orbital angular momentum beams, Adv. Opt. Photon. **7**, 66 (2015).

[2] M. Erhard, R. Fickler, M. Krenn, and A. Zeilinger, Twisted photons: new quantum perspectives in high dimensions, Light: Science & Applications **7**, 17146 (2018).

[3] B. C. dos Santos, K. Dechoum, and A. Z. Khoury, Continuous-variable hyperentanglement in a parametric oscillator with orbital angular momentum, Phys. Rev. Lett. **103**, 230503 (2009).

[4] K. Liu, J. Guo, C. Cai, S. Guo, and J. Gao, Experimental generation of continuous-variable hyperentanglement in an optical parametric oscillator, Phys. Rev. Lett. **113**, 170501 (2014).

[5] C. Cai, L. Ma, J. Li, H. Guo, K. Liu, H. Sun, R. Yang, and J. Gao, Generation of a continuous-variable quadripartite cluster state multiplexed in the spatial domain, Photon. Res. **6**, 479 (2018).

[6] R. W. Boyd, *Nonlinear optics* (Academic press, 2020).

[7] J. Wang, F. Castellucci, and S. Franke-Arnold, Vectorial light–matter interaction: Exploring spatially structured complex light fields, AVS Quantum Science **2**, 031702 (2020).

[8] R. F. Offer, A. Daffurn, E. Riis, P. F. Griffin, A. S. Arnold, and S. Franke-Arnold, Gouy phase-matched angular and radial mode conversion in four-wave mixing, Phys. Rev. A **103**, L021502 (2021).

[9] V. Kudriašov, M. M. Sinkevičienė, N. Daloi, J. Ruseckas, T. N. Dey, S. Franke-Arnold, and H. R. Hamedi, Propagation of optical vector and scalar vortices in an atomic medium with closed-loop tripod configuration, Opt. Express **33**, 40931 (2025).

[10] J. Wang, X. Yang, Y. Chen, Z. Ye, X. Zeng, Y. Zhou,

S. Zhang, C. M. Cisowski, C. Wang, K. Inoue, Y. Shen, S. Franke-Arnold, and H. Gao, Storage and retrieval of optical skyrmions with topological characteristics, Phys. Rev. A **113**, 043528 (2026).
[11] X. Pan, S. Yu, Y. Zhou, K. Zhang, K. Zhang, S. Lv, S. Li, W. Wang, and J. Jing, Orbital-angular-momentum multiplexed continuous-variable entanglement from four-wave mixing in hot atomic vapor, Phys. Rev. Lett. **123**, 070506 (2019).
[12] S. Li, X. Pan, Y. Ren, H. Liu, S. Yu, and J. Jing, Deterministic generation of orbital-angular-momentum multiplexed tripartite entanglement, Phys. Rev. Lett. **124**, 083605 (2020).
[13] K. Zhang, W. Wang, S. Liu, X. Pan, J. Du, Y. Lou, S. Yu, S. Lv, N. Treps, C. Fabre, and J. Jing, Reconfigurable hexapartite entanglement by spatially multiplexed four-wave mixing processes, Phys. Rev. Lett. **124**, 090501 (2020).
[14] Y. Chen, S. Liu, Y. Lou, and J. Jing, Orbital angular momentum multiplexed quantum dense coding, Phys. Rev. Lett. **127**, 093601 (2021).
[15] L. J. Pereira, W. T. Buono, D. S. Tasca, K. Dechoum, and A. Z. Khoury, Orbital-angular-momentum mixing in type-ii second-harmonic generation, Phys. Rev. A **96**, 053856 (2017).
[16] W. T. Buono, A. Santos, M. R. Maia, L. J. Pereira, D. S. Tasca, K. Dechoum, T. Ruchon, and A. Z. Khoury, Chiral relations and radial-angular coupling in nonlinear interactions of optical vortices, Phys. Rev. A **101**, 043821 (2020).
[17] B. P. da Silva, W. T. Buono, L. J. Pereira, D. S. Tasca, K. Dechoum, and A. Z. Khoury, Spin to orbital angular momentum transfer in frequency up-conversion, Nanophotonics **11**, 771 (2022).
[18] J. W. R. Tabosa and D. V. Petrov, Optical pumping of orbital angular momentum of light in cold cesium atoms, Phys. Rev. Lett. **83**, 4967 (1999).
[19] S. Barreiro and J. W. R. Tabosa, Generation of light carrying orbital angular momentum via induced coherence grating in cold atoms, Phys. Rev. Lett. **90**, 133001 (2003).
[20] A. J. F. de Almeida, S. Barreiro, W. S. Martins, R. A. de Oliveira, D. Felinto, L. Pruvost, and J. W. R. Tabosa, Storage of orbital angular momenta of light via coherent population oscillation, Opt. Lett. **40**, 2545 (2015).
[21] N. Šantić, A. Fusaro, S. Salem, J. Garnier, A. Picozzi, and R. Kaiser, Nonequilibrium precondensation of classical waves in two dimensions propagating through atomic vapors, Phys. Rev. Lett. **120**, 055301 (2018).
[22] R. S. N. Moreira, J. C. de Aquino Carvalho, and J. W. R. Tabosa, Amplification and storage of light carrying orbital angular momentum via magnetically assisted gain in cold atoms, Journal of Optics **25**, 024003 (2022).
[23] M. R. L. da Motta, A. A. C. de Almeida, and S. S. Vianna, Spatial distribution of two symmetric four-wave-mixing signals induced by gaussian beams, Phys. Rev. A **106**, 053502 (2022).
[24] M. R. L. da Motta, A. A. C. d. Almeida, and S. S. Vianna, Combinations of orbital angular momentum in two degenerate four-wave mixing processes in rb vapor, Journal of Optics **25**, 095501 (2023).
[25] M. R. L. da Motta, G. B. Alves, A. Z. Khoury, and S. S. Vianna, Poincaré-sphere symmetries in four-wave mixing with orbital angular momentum, Phys. Rev. A **109**, 013506 (2024).
[26] T. Boulier, E. Cancellieri, N. D. Sangouard, Q. Glorieux, A. V. Kavokin, D. M. Whittaker, E. Giacobino, and A. Bramati, Injection of orbital angular momentum and storage of quantized vortices in polariton superfluids, Phys. Rev. Lett. **116**, 116402 (2016).
[27] S. Donati, L. Dominici, G. Dagvadorj, D. Ballarini, M. D. Giorgi, A. Bramati, G. Gigli, Y. G. Rubo, M. H. Szymańska, and D. Sanvitto, Twist of generalized skyrmions and spin vortices in a polariton superfluid, Proceedings of the National Academy of Sciences **113**, 14926 (2016), https://www.pnas.org/doi/pdf/10.1073/pnas.1610123114.
[28] S. V. Koniakhin, O. Bleu, D. D. Stupin, S. Pigeon, A. Maitre, F. Claude, G. Lerario, Q. Glorieux, A. Bramati, D. Solnyshkov, and G. Malpuech, Stationary quantum vortex street in a driven-dissipative quantum fluid of light, Phys. Rev. Lett. **123**, 215301 (2019).
[29] G. Lerario, A. Maître, R. Boddeda, Q. Glorieux, E. Giacobino, S. Pigeon, and A. Bramati, Vortex-stream generation and enhanced propagation in a polariton superfluid, Phys. Rev. Res. **2**, 023049 (2020).
[30] K. Guerrero, K. Falque, E. Giacobino, A. Bramati, and M. J. Jacquet, Multiply quantized vortex spectroscopy in a quantum fluid of light, Phys. Rev. Lett. **135**, 243801 (2025).
[31] H.-J. Wu, L.-W. Mao, Y.-J. Yang, C. Rosales-Guzmán, W. Gao, B.-S. Shi, and Z.-H. Zhu, Radial modal transitions of laguerre-gauss modes during parametric up-conversion: Towards the full-field selection rule of spatial modes, Phys. Rev. A **101**, 063805 (2020).
[32] A. de Oliveira, G. Santos, N. R. da Silva, L. Pereira, G. Alves, A. Khoury, and P. S. Ribeiro, Beyond conservation of orbital angular momentum in stimulated parametric down-conversion, Phys. Rev. Appl. **16**, 044019 (2021).
[33] V. V. Kotlyar, E. G. Abramochkin, A. A. Kovalev, and A. A. Savelyeva, Product of two laguerre-gaussian beams, Photonics **9**, 10.3390/photonics9070496 (2022).
[34] E. V. G. Ramirez, M. L. A. Carrasco, M. M. M. Otero, S. C. Cerda, and M. D. I. Castillo, Far field intensity distributions due to spatial self phase modulation of a gaussian beam by a thin nonlocal nonlinear media, Opt. Express **18**, 22067 (2010).
[35] F. Dalfovo, S. Giorgini, L. P. Pitaevskii, and S. Stringari, Theory of bose-einstein condensation in trapped gases, Rev. Mod. Phys. **71**, 463 (1999).
[36] J. Mendonça, R. Bingham, and P. Shukla, A kinetic approach to bose-einstein condensates: Self-phase modulation and bogoliubov oscillations, J. Exp. Theor. Phys. **101**, 942 (2005).
[37] Q. Glorieux, Chapter four - paraxial fluids of light, in *Paraxial fluids of light*, Advances In Atomic, Molecular, and Optical Physics, Vol. 74, edited by L. F. Dimauro, H. Perrin, and S. Yelin (Academic Press, 2025) pp. 157–241.
[38] M. G. de Oliveira, L. J. Pereira, A. S. Santos, K. Dechoum, A. Bramati, and A. Z. Khoury, Radial-angular coupling in self-phase-modulation with structured light, Phys. Rev. A **108**, 013503 (2023).
[39] L.-M. Duan, G. Giedke, J. I. Cirac, and P. Zoller, Inseparability criterion for continuous variable systems, Physical review letters **84**, 2722 (2000).
[40] P.-É. Larré and I. Carusotto, Propagation of a quantum

fluid of light in a cavityless nonlinear optical medium: General theory and response to quantum quenches, Phys. Rev. A **92**, 043802 (2015).

[41] P. Hyllus and J. Eisert, Optimal entanglement witnesses for continuous-variable systems, New Journal of Physics **8**, 51 (2006).

[42] N. Treps and C. Fabre, Transverse distribution of quantum fluctuations and correlations in spatial solitons, Phys. Rev. A **62**, 033816 (2000).

[43] E. Lantz, T. Sylvestre, H. Maillotte, N. Treps, and C. Fabre, Quantum fluctuations and correlations of spatial scalar or multimode vector solitons in kerr media, Journal of Optics B: Quantum and Semiclassical Optics **6**, S295 (2004).

[44] Y. Lai and S.-S. Yu, General quantum theory of nonlinear optical-pulse propagation, Phys. Rev. A **51**, 817 (1995).

[45] R.-K. Lee and Y. Lai, Quantum theory of fibre bragg grating solitons*, Journal of Optics B: Quantum and Semiclassical Optics **6**, S638 (2004).